Cross-Lingual Embedding Alignment for Six Ancient Languages: Pre-Registered Findings from the 2026-07 Arc

hyper-glyphy research program

2026-07-24

Contents

Abstract

We report the results of one research arc of the hyper-glyphy project, which aligns monolingual embeddings for six ancient languages — Sumerian, Egyptian, Akkadian, Hittite, Greek, and Sanskrit — into shared English target spaces. The arc rests on an honest-evaluation reset (2026-07-06) that invalidated all prior headline numbers as surface-variant leakage artifacts and replaced them with a stratified CSLS suite under a lemma-group split. On that foundation we ran a pre-registered anchor-quality experiment: the Sanskrit slot, built on the strongest lexicon join in the project (94.9%), landed its Procrustes validation cosine at 0.1145, inside the pre-registered ≤ 0.12 band — anchors were never the constraint, and the stronger-anchors lever and Plane A were retired. Suite v2 (shared gloss filters, plateau alpha rule) then improved combined test accuracy in all six slots on the primary whitened-Gemma target, fixed the Akkadian-Gemma alpha-selection pathology, and left the whitened-Gemma target ahead of GloVe in six of six slots. The capstone $K=5$ myth study delivered the first adequately-powered Plane-B test in the study's history — a real null (sumerian–sanskrit, $\rho=-0.33$, exhaustive $p=0.77$) — alongside the study's first pre-registered positive, a band-edge result for Vṛtra↔Illuyanka (90.55th percentile) that carries an explicit midrank/tie-block caveat. Every one of these verdicts was banded before measurement; the four verdict sentences are reproduced verbatim from the experiment journal.

1 Introduction

The hyper-glyphy project asks a narrow, testable question: how much of the lexical and thematic structure of an ancient language survives a linear map from a monolingual embedding space into a modern English semantic space? Six language slots — Sumerian, Egyptian, Akkadian, Hittite, Classical Greek, and Sanskrit — share one pipeline: FastText embeddings trained per language, fused to a common dimensionality, and aligned by Ridge regression into two English targets (GloVe 300d and a whitened EmbeddingGemma 768d space). Three of the six slots are written in cuneiform script (the divine determinative 𒀭[1] will be familiar from any Sumerian or Hittite tablet); the others span hieroglyphic transliteration, polytonic Greek, and IAST-romanized Sanskrit.

This paper covers one arc of work, July 2026, and its through-line is methodological: **pre-registration as method**. Every major claim in this arc — the anchor-quality verdict, both myth-study read-outs, and the document-level gates — was banded before measurement, and each verdict sentence below is byte-copied from the experiment journal entry that recorded it. Where a result sits at the edge of its band, the paper says so in the same breath.

The arc's foundation is the eval-integrity reset of 2026-07-06. A repo-wide review found two structural flaws in every slot's evaluation: surface-variant train/test leakage — anchor extraction deliberately registers multiple surfaces per lemma with the same gloss, and a random split over *pairs* let (šarrum, "king") train while (šarru, "king") tested — and alpha tuned on the test set. Measured on the exact shipped splits, the fraction of test items with a same-gloss train anchor within edit distance 1 ranged from 29.5% (Egyptian) to 65.2% (Greek, projected). The fix — a shared union-find lemma-group split (64/16/20 train/val/test) with alpha selected on validation — collapsed the Akkadian rerun from 27.79% to 0.09% top-1 (GloVe) and from 36.43% to 0.14% (whitened Gemma). A three-way diagnostic (44.4% accuracy on the model's own training anchors; a near-reproduction of the old number — 27.59% vs 27.79% shipped — when the new code is re-run under the old random-pair split, quantifying the leakage; only

[1] Cuneiform sign provenance, per the Anomaly Atlas convention: U+1202D, Unicode character name CUNEIFORM SIGN AN — the divine determinative, standard in ORACC cuneiform encoding.

16.3% of lemma-split test glosses appearing anywhere in train) confirmed the collapse was real zero-shot difficulty, not a bug. All prior headline numbers were declared invalidated, and every number in this paper post-dates that reset.

Two further disciplines carried through the arc. First, a stratified evaluation (Section 2.3) that separates memorization from generalization instead of averaging them. Second, disclosure: every deviation from a shared recipe, every user-accepted trade-off, and every band-edge caveat is recorded in the journal at the moment of measurement, and reproduced here rather than smoothed over.

The arc's four results, in order: (i) suite v2, a cleanup pass (shared gloss filters, a plateau alpha rule) that improved combined accuracy in all six slots on the primary whitened-Gemma target and fixed a known selection pathology (Section 3.1); (ii) the Sanskrit slot as a pre-registered anchor-quality experiment whose verdict retired an entire improvement lever (Section 3.2); (iii) the document-level gates that routed the myth study away from cross-language cosine (Section 3.3); and (iv) the K=5 myth study, the first adequately-powered test of cross-language thematic geometry in this project, returning a real null beside a band-edge positive control (Section 3.4).

2 Data and methods

2.1 Language slots

Table 1 summarizes the six slots. Corpus and lexicon facts are taken from the per-slot data manifests (`languages/<slot>/README.md` and `languages/<slot>/data/raw/README.md`).

Table 1: The six language slots. Anchor counts are suite-v2 values (journal, 2026-07-19); the Egyptian count is the 06-output basis, not directly comparable to its v1 post-filter figure (4,152).

Slot	Family / script	Corpus (source, size)	Lexicon (anchor source)	Anchors (v2)
Sumerian	language isolate / cuneiform	ETCSL (36K lines) + CDLI (96K texts, 1.4M lines) + ORACC (90K texts, 4.3M lemmas)	ePSD2 glosses + ETCSL co-occurrence	13,048
Egyptian	Afroasiatic / hieroglyphic	migrated pre-built from the predecessor heiroglyphy V15 project	TLA / Ramses / BBAW pairs (8,541 raw)	4,568
Akkadian	East Semitic / cuneiform	ORACC OB + SB projects + letters + DCCLT, 3.0M tokens	ORACC guide-word glosses	24,116
Hittite	Indo-European (Anatolian) / cuneiform	TLHdig 0.2.0-beta (Zenodo, 22k XML texts, CC-BY)	German guide words, translated via multilingual EmbeddingGemma; heterogram bridge (sux 28 / akk 39)	11,651

Slot	Family / script	Corpus (source, size)	Lexicon (anchor source)	Anchors (v2)
Greek	Indo-European / alphabetic	Diorisis v1.51 (821 JSON files, ~10.2M tokens, Homer to 5th c. AD)	Perseus LSJ (90,424 entries; join hit rate 77.1%)	105,920
Sanskrit	Indo-European (Indo-Aryan) / Devanagari (IAST)	DCS (15,790 chapter files, 5,679,462 token-lemma records, 90,184 unique lemmas; CC BY 4.0)	Monier-Williams, Cologne CDSL 2020 (177,323 entries; CC BY-NC-SA 3.0; join hit rate 94.9%)	92,275

2.2 Pipeline

Each slot trains 768-dimensional FastText skip-gram embeddings on its cleaned corpus, zero-pads them to a fused 1536-dimensional representation, and learns one Ridge regression per target space: (a) whitened EmbeddingGemma 768d — BERT-whitening (Su et al. 2021) applied to the raw encoder output, which earlier work found mandatory for any contextual-encoder alignment target — and (b) GloVe 6B 300d. Both aligned views are exposed through per-language `Lookup` classes (`space="gemma"|"glove"`).

2.3 Word-level evaluation suite

All word-level numbers use the stratified CSLS suite introduced 2026-07-09. Anchors are split by a shared union-find lemma-group procedure (anchors merge on shared lemma or shared surface; gloss fallback for Egyptian, which has no lemma key), 64/16/20 train/val/test, seed 42, with near-surface edges merged so that no test item has a same-gloss train anchor within edit distance 1 (leak check 0.00% in all slots). Retrieval is CSLS (k=10 mutual-neighbor reranking) over a 50,000-candidate English vocabulary, reporting exact-match and WordNet-synonym-credited top-1/5/10. Three regimes are reported per slot: **dictionary** (a fixed 1,000-anchor sample of the training set — in-sample by design, measuring memorization of known glosses), **interpolation** (unseen test lemmas whose gold gloss appears as a training target), and **zero-shot** (test lemmas whose gold gloss was never a training target). Headline accuracies are conditioned on the gold gloss being present in the 50K candidate vocabulary; a **gold-OOV** column reports how many test items that restriction excludes (shown as excluded/evaluated).

Suite v2 (2026-07-19) made two recipe changes, holding the eval data fixed. First, a shared `gloss_filters` module became the single source of anchor-English selection across all six slots: negator rejection (*not, no, without, never*, plus German equivalents for Hittite), cross-reference-starter rejection (*see, cf, vid*), a 14-word scaffold-word list, and a Unicode word regex, with a MIN_HIT_RATE 0.40 gate unchanged from v1. Second, alpha selection moved to a plateau rule (**alpha-v2**): select on validation top-5 CSLS, define the plateau as within 100/n_val percentage points of the maximum, and break ties toward the lowest alpha (v1 selected on val top-1 with an arbitrary tie-break). All twelve results configurations (6 slots × 2 targets) record `alpha_selection=val_top5_csls_v2`.

2.4 The Procrustes anchor-quality read-out

To test whether anchor quality — rather than map family or corpus scale — was the binding constraint on cross-language document geometry, each slot also fits a semi-orthogonal Procrustes map ($W = UV^T$ of X^TY, 1536→768, no scale), with full and stable-monosemous anchor variants selected on validation cosine. The isometry check matters here: the fused 1536d embeddings have exact rank 768 (FastText 768d plus a projected component), so a 1536→768 semi-orthogonal map can capture the data subspace exactly (isometry $\rho \approx 1.0$, no projection loss), and a low read-out cannot be blamed on the projection.

The read-out was pre-registered in the 2026-07-13 Sanskrit design spec, before the Sanskrit slot was built. Bands, restated: validation cosine $\geq 0.20 \Rightarrow$ anchor quality was a binding constraint and the stronger-anchors lever stays live; $\leq 0.12 \Rightarrow$ anchors were never the constraint; between 0.12 and 0.20 $\Rightarrow$ inconclusive. Sanskrit was built to be the stronger-anchors condition: its token-level DCS-lemma→MW join hit rate of 94.9% is the best of any slot (the others' joins are substantially thinner).

2.5 Myth study design

The myth study asks whether cross-language *thematic* geometry — not word-level translation — is detectable between the document collections of different slots. Two planes were defined in the study plan. **Plane A** is direct cross-language cosine in the aligned spaces; it was gated on two document-level criteria (Section 3.3). **Plane B** is native-space second-order representational similarity analysis (RSA): within each slot, theme-centroid profiles are computed from the slot's own fused embeddings (no cross-language cosine is ever taken), and the *structure* of those profiles is compared across slots by Spearman correlation of the theme-pair similarity ladders, with exhaustive permutation tests.

Five themes are defined — cosmogonic, hymnic, wisdom, royal_control, magical — with five documents per filled theme per slot. Greek lacks a magical theme (the Greek Magical Papyri are not in the Diorisis literary corpus) and Hittite lacks wisdom; a slot pair's ladder uses only shared themes, so ladder size K ranges from 3 to 5. Power is set by K: a K=3 ladder yields 3 upper-triangle values and 6 exhaustive permutations (minimum attainable $p = 0.1667$ — structurally unpowered at $\alpha = 0.05$); K=4 yields 24 permutations (min $p \approx 0.042$); K=5 yields 120 (min $p \approx 0.008$). The Sanskrit slot is the only one that fills all five themes, making sumerian–sanskrit the study's first K=5 ladder. The Sanskrit roster (DCS chapter-level documents, pinned 2026-07-23) comprises Ṛgveda cosmogonic hymns (RV 10.129, 10.90, 10.121, 10.190, plus a merged Vṛtra document = RV 1.32 + 1.80 + 2.12, 862 tokens), the five longest non-cosmogonic RV hymns, five principal Upaniṣads as wisdom documents, five Atharvaveda royal-consecration chapters, and five long Atharvaveda incantation chapters (excluding, by rule, the wedding and funerary books 14 and 18).

Two rules were pre-registered before the run. **Read-out 1 (K=5 ladder RSA):** if the shared ladder reaches its maximum, exhaustive permutation $p \leq 0.05$ with positive Spearman ρ counts as the study's first adequately-powered Plane-B positive, stated as such; $p > 0.05$ is a real, reportable null, stated as such. **Read-out 2 (Vṛtra positive control):** the Vṛtra profile correlation's percentile in a 1000-draw same-genre null at $\geq$ the 90th percentile $\Rightarrow$ "supports the IE combat-myth link"; $\leq$ the 75th $\Rightarrow$ "fails, consistent with the Kumarbi-control finding"; between $\Rightarrow$ inconclusive. The earlier doc-level Kumarbi control (Hittite succession narratives vs the Theogony, 2000-draw null) carries the same band logic and serves as the reference point.

3 Results

3.1 Word-level suite v2

Table 2 reports the current (suite v2) word-level numbers for all six slots and both targets.

Table 2: Suite v2 (2026-07-19), top-1 exact % per stratum plus synonym-credited combined and the gold-OOV column (excluded/evaluated). † Akkadian Gemma v1's alpha=1e4 flat-noise pick (dict 19.85%) is fixed under v2's plateau rule (see below). ‡ Hittite zero-shot n=0 hits at both alphas; gold OOV 850/419 — the majority of Hittite test items are OOV of the 50K candidate vocabulary. § Sanskrit Gemma alpha=1e4 is a real interior signal max, not a flat-noise pick (user-accepted trade, 2026-07-18; see text).

Slot	Target	alpha	Dict top-1	Interp top-1	Zero-shot top-1	Combined top-1	Combined syn	Gold OOV
Sumerian	GloVe	10	77.99%	7.20%	0.20%	4.44%	5.32%	145/1240
Sumerian	Gemma	1000	75.60%	10.13%	1.22%	6.61%	7.42%	145/1240
Egyptian	GloVe	1e-3	66.29%	25.15%	0.00%	19.07%	19.73%	87/451
Egyptian	Gemma	1e-3	70.52%	26.02%	0.92%	19.96%	21.51%	87/451
Akkadian	GloVe	10	38.47%	0.00%	0.24%	0.18%	0.69%	95/2184
Akkadian	Gemma	0.01 †	55.24%	0.18%	0.31%	0.27%	0.55%	95/2184
Hittite	GloVe	1e-4	67.99%	8.90%	0.00% ‡	5.01%	5.01%	850/419
Hittite	Gemma	1e-3	78.35%	13.98%	0.00% ‡	7.88%	8.59%	850/419
Greek	GloVe	0.1	37.10%	4.43%	0.59%	3.60%	6.24%	2145/15632
Greek	Gemma	1.0	50.11%	6.07%	0.68%	4.91%	8.12%	2145/15632
Sanskrit	GloVe	1e-4	35.16%	3.47%	0.09%	2.95%	4.79%	1331/14217
Sanskrit	Gemma	1e4 §	35.90%	5.49%	0.46%	4.71%	7.28%	1331/14217

Three observations. First, **combined test accuracy on the primary whitened-Gemma target improved in all six slots** relative to suite v1 (Egyptian 15.47% → 19.96%, the biggest v2 winner; Sumerian 5.54% → 6.61%; Akkadian 0.13% → 0.27%; Hittite 6.83% → 7.88%; Greek 4.37% → 4.91%; Sanskrit 3.83% → 4.71%). The anchor-count guardrail passed in all six slots; v2's stricter filters reject between 52 (Sumerian) and 3,649 (Sanskrit, −3.8%) of the raw joins.

Second, the **memorization/generalization trade** is now explicit rather than hidden in an average. The dictionary stratum (in-sample memorization) runs 35–78% top-1 across slots at 50K candidates, while zero-shot remains near zero everywhere (0.00–1.22%): the linear map retrieves known glosses well and does not generalize to unseen lemma families at these corpus scales. Sanskrit's Gemma alpha illustrates the trade directly: alpha=1e4 is a genuine interior maximum of the validation signal that gives up in-sample dictionary accuracy (44.40% v1 → 35.90% v2) in exchange for improvement in every test stratum — a trade accepted deliberately and recorded as such (journal, 2026-07-19).

Third, the **Akkadian-Gemma pathology is fixed, not papered over**. This was the pre-registered acid test for the alpha-v2 rule: v1's Gemma dictionary accuracy of 19.85% came from alpha=1e4, a flat-noise pick that won by a single anchor over an entirely flat 0.1–1000 validation plateau. The plateau rule, applied to the *same* flat sweep, ties toward the lowest alpha and picks 0.01 — dictionary jumps to 55.24% and combined from 0.13% to 0.27%, with no change to the eval data (Figure 1). With this fix, the whitened-Gemma target beats GloVe on combined accuracy in **six of six slots**.

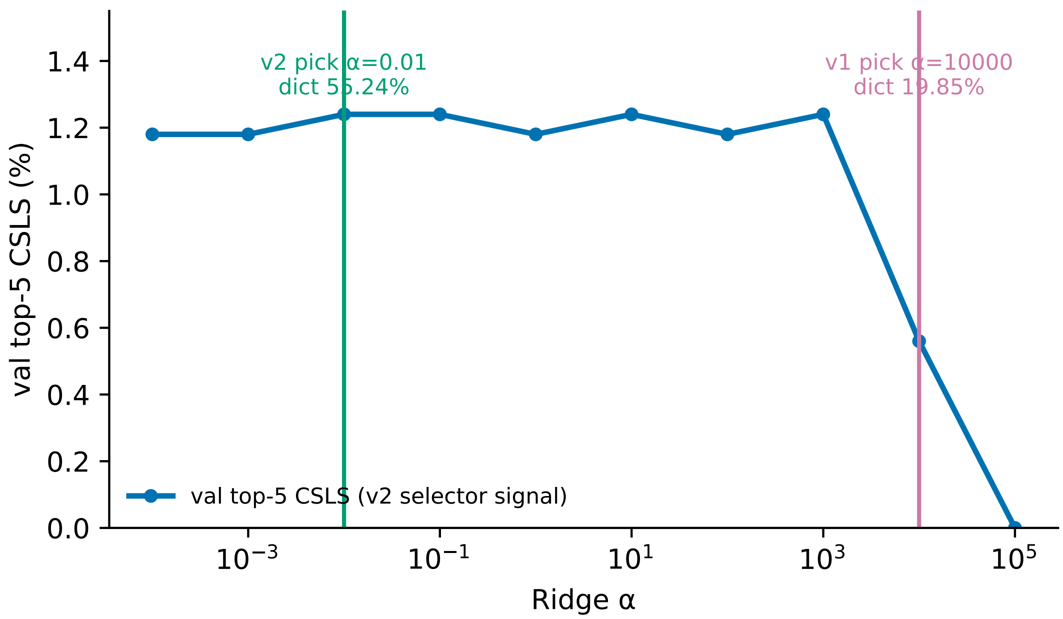

Figure 1: Akkadian Gemma validation sweep with the v1 top-1 pick (α=10^4, a flat-noise selection; dictionary 19.9%) and the v2 plateau pick (α=0.01; dictionary 55.2%). Same sweep, same eval data — only the selection rule changed.

3.2 The anchor-quality experiment: Sanskrit and the retirement verdict

The Sanskrit slot (shipped 2026-07-16) was built to answer a standing question with a pre-registered read-out: would a stronger anchor set move the Procrustes plane? The slot's inputs are the strongest in the project — DCS parsed with 0.000% loss (0 of 6,713,257 token lines), a 94.9% token-level lemma→lexicon join against Monier-Williams (5,391,784 hits / 287,678 misses), 95,924 v1 anchors with 90,176 valid at fit time. Its word-level v1 suite behaved like the other slots' (Gemma combined 3.83%, beating GloVe's 2.22%; dictionary 44.40%), i.e. a normal member of the family rather than an outlier.

The Procrustes read-out came back at validation cosine **0.1145** (full variant 0.11451, n=57,834, chosen over stable 0.11422; n_fit 70,324; isometry ρ 1.0000). That falls in the pre-registered **≤ 0.12** band, and the pre-registered verdict was applied verbatim per the spec:

> "anchors were never the constraint ⇒ retire the stronger-anchors lever, and with it the last named route to Plane A."

The reference slots make the verdict legible (Figure 2): sumerian 0.1157, greek 0.1149, hittite 0.0586, sanskrit 0.1145. Three slots with very different corpus and lexicon quality — Sumerian's composite cuneiform corpus, Greek's 77.1% LSJ join, Sanskrit's 94.9% MW join — converge at ~0.115, which suggests a structural ceiling on the semi-orthogonal plane rather than an anchor-quality effect; Hittite's lower 0.0586 remains unexplained by this test. Suite v2's re-measurement (recorded for the record, not re-litigated: sanskrit 0.1198, sumerian 0.1117, greek 0.1163, hittite 0.0666) left all four slots ≤ 0.12; the v1 retire verdict was pre-registered on the v1 recipe and stands. Sanskrit's v2 value, 0.1198, sits closest of the four to the 0.12 band edge — worth noting factually.

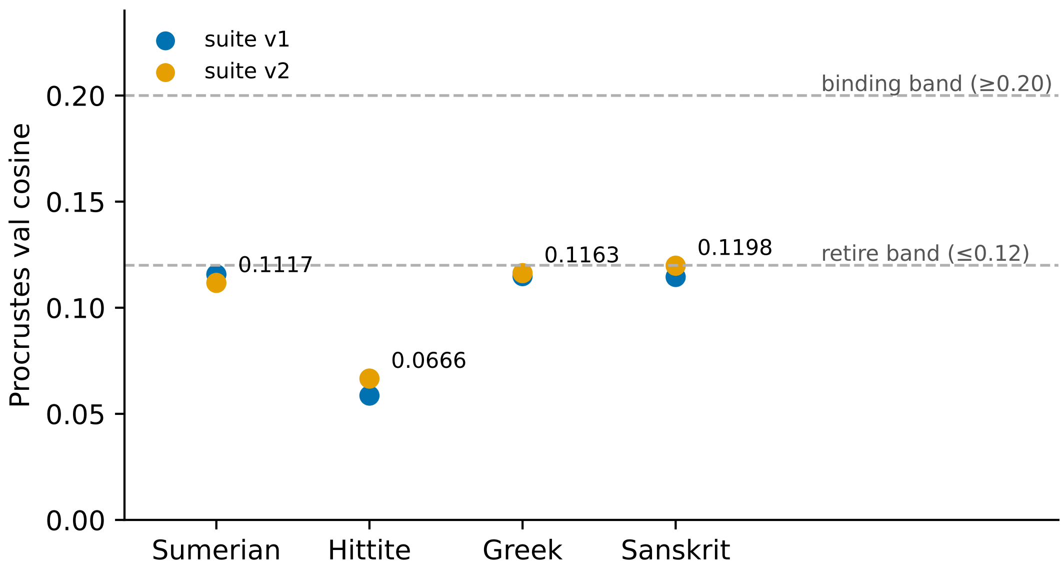

Figure 2: Procrustes convergence: per-slot validation cosines under the v1 and v2 recipes against the pre-registered bands (retire ≤ 0.12, binding ≥ 0.20). Sumerian, Greek, and Sanskrit converge at ~0.115 despite very different anchor quality; no slot approaches the 0.20 band.

3.3 Document-level gates

Two document-level panels gate cross-language claims. **Gate 1** (within-language): Sumerian ETCSL genre leave-one-out over 338 compositions, 5 genres, majority baseline 40.83% — gemma_aligned 63.31% (+22.5 pp), glove_aligned 60.95% (+20.1 pp), fused_unaligned 68.05% (+27.2 pp; projection cost ≈ 5 pp). The criterion (≥ 15 pp over majority in at least one aligned space) is a **PASS**: the aligned spaces retain real within-language document structure.

Gate 2 (cross-language): parallel retrieval, Hittite → Greek over a pool of 820 Greek documents, with the Kumarbi succession narrative vs Hesiod's Theogony as positive control. Criterion: MRR ≥ 0.1 with the positive control in the top quartile. Measured on the Ridge plane: Kumarbi→Theogony rank 731/820, Illuyanka→Theogony 781/820, Ullikummi→Theogony 788/820, MRR 0.0013 — **FAIL**, with all three pairs in the bottom 11% of the pool. Word-level alignment does not compose into cross-slot document retrieval. Re-measured on the semi-orthogonal Procrustes plane (2026-07-13): ranks 526/798/796, MRR 0.0015 — FAIL again; Plane A stays no-go under both map families.

A follow-up hubness diagnostic (2026-07-19, suite-v2 space) tested the anti-hub hypothesis for the Gate-2 failure and returned a **partial** verdict. All three control pairs target the same Greek document (the Theogony), whose SIF centroid is a genuine norm outlier — L2 norm 4.02 against a pool mean of 4.69 (std 0.32), the 1.8th percentile, suggesting near-cancellation from lexical diversity. But its mean cosine to the rest of the pool sits only at the 15th percentile (0.853 vs pool mean 0.885) — below median, not bottom-decile — and ranks recomputed in the v2 space (340/770/733) are better than v1's, with Kumarbi→Theogony no longer bottom-decile at all. The Theogony carries a real geometric anomaly, but not a clean "far from everything" anti-hub signature; some pair-specific factor remains unaccounted for.

3.4 The myth study at K=5

With Plane A gated off, the myth study runs entirely on Plane B — native-space second-order RSA — and suite v2 supplied the spaces for a full re-run with Sanskrit as the fourth slot (2026-07-24; six slot pairs, zero dropped documents; every pinned roster ID resolved). Table 3 and Figure 3 give all six pair RSAs.

Table 3: All six slot-pair ladder RSAs (exhaustive permutation tests). The sumerian–sanskrit pair is the study's first K=5 ladder.

Pair	K	ρ	p	n_perms	Themes dropped
sumerian–hittite	4	0.3143	0.25	24	wisdom
sumerian–greek	3	1.0	0.1667	6	wisdom, magical
sumerian–sanskrit	**5**	**−0.3333**	**0.7667**	**120**	none
hittite–greek	3	0.5	0.5	6	magical
hittite–sanskrit	4	−0.3714	0.8333	24	wisdom
greek–sanskrit	3	−0.5	0.8333	6	wisdom, magical

Read-out 1. Bands, restated: at the ladder maximum, exhaustive $p \leq 0.05$ with positive ρ would have been the study's first adequately-powered Plane-B positive; $p > 0.05$ is a real, reportable null. Measured: sumerian–sanskrit K=5, ρ=−0.3333, exhaustive p=0.7667 (n_perms=120) — $p > 0.05$, verdict verbatim:

> "a real, reportable null — thematic geometry does not detectably align."

This is the first test in the study with enough permutations to clear $\alpha = 0.05$, and it comes back negative-signed and null. The hittite–sanskrit K=4 ladder (ρ=−0.3714, p=0.8333) is also null under the same rule. The IE gradient does not rescue a relatedness story: the mean ρ over the three Indo-European pairs is −0.1238 against 0.327 for the non-IE pairs — both Sanskrit IE pairs are negative, so adding the IE triangle's third leg does not elevate the gradient; if anything the non-IE pairs read higher on this measurement.

Read-out 2. Bands, restated: Vṛtra profile-correlation percentile in the 1000-draw same-genre null ≥ 90th ⇒ supports; ≤ 75th ⇒ fails; between ⇒ inconclusive. Measured (Figure 4): vs Illuyanka (K=4 ladder), ρ=0.4, percentile 90.55 ⇒ verdict verbatim:

> "supports the IE combat-myth link"

— the study's first pre-registered positive, and on the phylogenetically closest pair (both Indo-European, both featuring a storm-god/serpent combat myth). Stated plainly, as the journal states it: this is one sub-control clearing the band by a narrow margin (90.55 vs the 90th-percentile line), not a strong positive. Vs Theogony (K=3 ladder), ρ=0.5, percentile 62.3 ⇒ verdict verbatim:

> "fails, consistent with the Kumarbi-control finding."

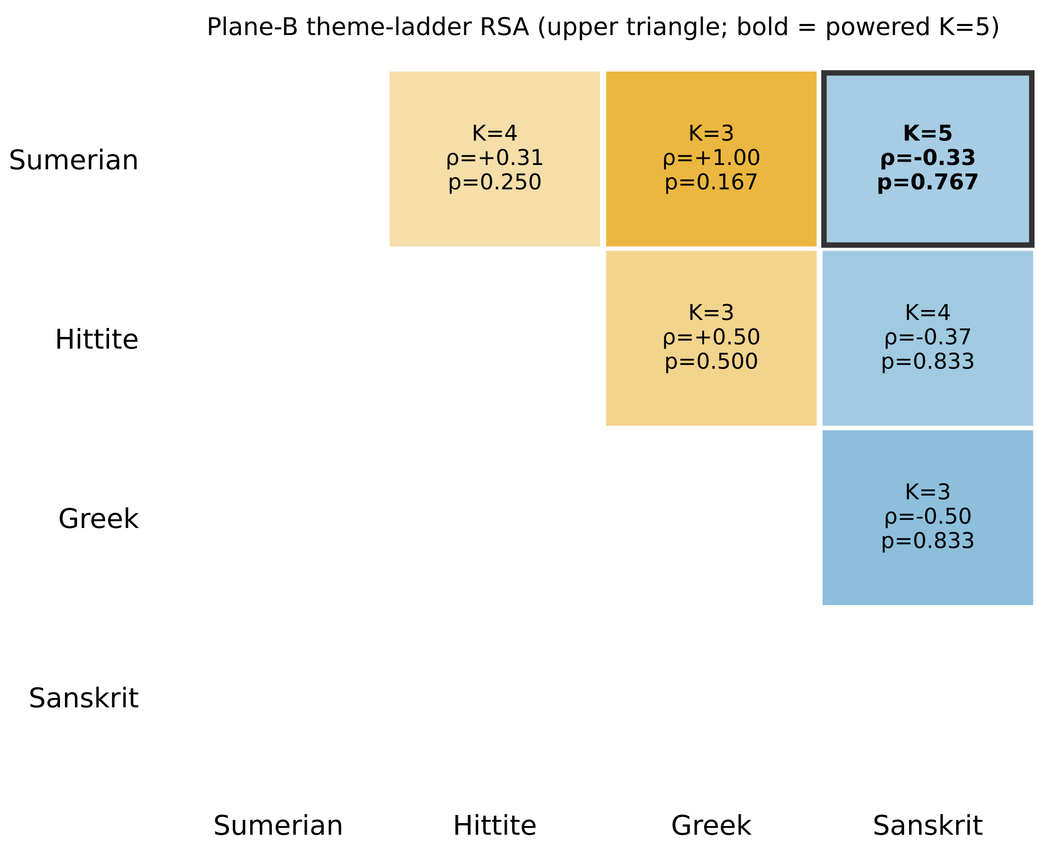

Figure 3: Slot-pair RSA matrix: shared-ladder K, Spearman ρ, and exhaustive permutation p for the six pairs. The sumerian–sanskrit cell is the study's first adequately-powered (K=5) test.

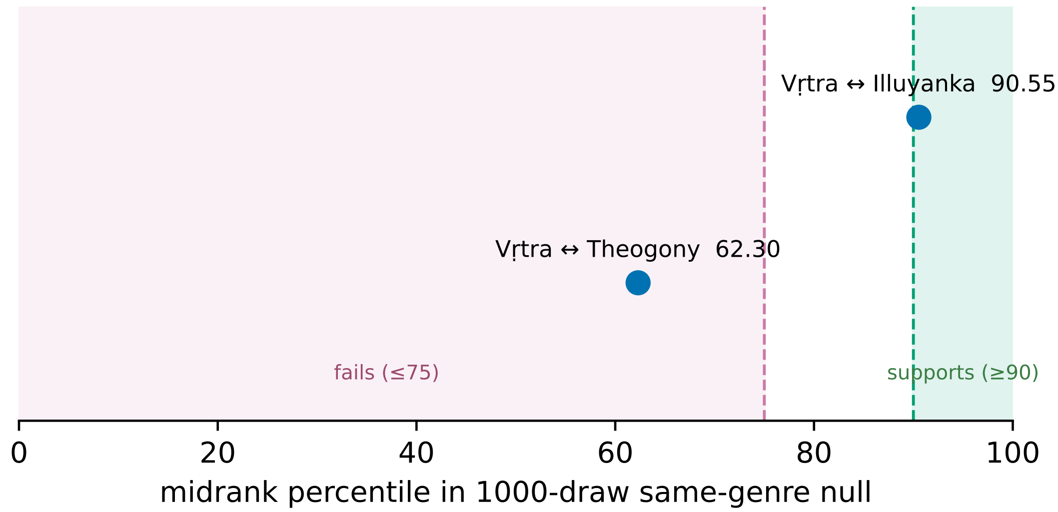

Figure 4: Vṛtra positive control: observed profile-correlation percentile against the 1000-draw same-genre null for both sub-controls, with the pre-registered 90th/75th-percentile bands. Vṛtra↔Illuyanka (ρ=0.4) clears the support band at the 90.55th percentile; Vṛtra↔Theogony (ρ=0.5) lands at the 62.3rd and fails. Caveat, quoted from the journal: "the 90.55th percentile is a midrank value: the .55 margin itself reflects null draws tying the observed ρ=0.4 (the tie mass at the observed value is not separately recorded in this run), so the clearance over the 90th-percentile band is within one tie-block's width. The pre-registered verdict is defined on the midrank percentile and stands; the discreteness caveat applies with full force."

For that ladder, ~75.4% of the null ties at ρ=+0.5 — the Vṛtra-vs-Theogony profile sits exactly at the null's tied mode, the same pattern the Kumarbi control showed at K=3.

Kumarbi control stability. The doc-level positive control (Kumarbi/Ullikummi/Illuyanka vs Theogony, 2000-draw null) is pure Plane B — it runs entirely on native fused embeddings, which suite v2 did not touch — and re-running it on the v2 spaces reproduced the v1 numbers bit-for-bit: kumarbi–theogony ρ=0.5, percentile 58.98; ullikummi–theogony ρ=0.5, 58.98; illuyanka–theogony ρ=1.0, 90.15. Both named plan controls sit at the null's 42.6%-tied mode; the non-plan-named Illuyanka pair reaches the discreteness ceiling at ρ=+1.0 (~19.7% of the null tied there). Verdict, unchanged and verbatim: "FAIL — named controls at null mode; per plan, cross-language doc-level claims narrow to within-language planes." The bit-for-bit reproduction is itself informative: the Kumarbi result is a property of the native embeddings, not an artifact of the (now-changed) Ridge recipe.

Supporting measurements. The translation delta (Spearman correlation between native-space and aligned-space document-distance rankings, per slot) is high everywhere: Sumerian 0.8838 (39 docs), Hittite 0.8514 (18), Greek 0.9044 (12), Sanskrit 0.9378 (25, the highest of the four) — aligning distorts within-language document geometry only mildly. Cosmogonic concept fingerprints (within-language centroid profiles over ten concepts, compared across slots in aligned space) are all measurable (`fingerprint_status` ok in all four slots); the six cross-slot correlations run from sumerian–greek 0.8788 down to hittite–sanskrit 0.6242, but per the results JSON's baseline note, all must be read against the mismatched-theme cross-slot baseline (pair means 0.55–0.85, maxima up to 0.96) — none of the six is dramatically elevated above its own pair's mismatched baseline.

4 Discussion and limitations

What the maps support — and what they cannot do. The arc's positive findings are all within-language: aligned spaces classify Sumerian genre at +22.5 pp over the majority baseline, preserve document geometry at Spearman 0.85–0.94 against the native spaces, and retrieve memorized glosses at 35–78% top-1 over 50K candidates. Everything cross-language failed or nulled at adequate power: cross-slot document retrieval is worse than chance for the known-parallel controls under both map families; the first K=5 RSA ladder is a signed-negative null; and the matched-theme fingerprint correlations do not separate from their mismatched baselines. The consistent picture is that these linear maps transport lexical memorization and coarse within-language structure, and do not carry cross-language thematic geometry at the scales measured. The Procrustes convergence of three dissimilar slots at ~0.115 says the constraint is structural, not a fixable input-quality problem — which is exactly why the pre-registered experiment was allowed to retire the lever.

Band-edge honesty. The arc's single pre-registered positive clears its band by 0.55 percentile points, and that margin is narrower than it looks: the 90.55th percentile is a midrank value, so the clearance is within one tie-block's width of the 90th-percentile line (the tie mass at the observed ρ=0.4 was not separately recorded in this run). The verdict is defined on the midrank percentile and stands; a reader should nevertheless weight it as a band-edge result, supported by one sub-control while the other (vs Theogony) fails.

OOV conditioning. All headline accuracies are conditioned on the gold gloss being inside the 50K-candidate vocabulary. This is disclosed per slot (Table 2's gold-OOV column) and is severe for Hittite, where 850 test items are excluded and only 419 evaluated — a consequence of German→English gloss translation producing specialized vocabulary. Combined accuracies are therefore not comparable across slots without reading that column.

Ladder discreteness and power. Spearman ρ over K=3 ladders takes only the values $\{\pm1, \pm0.5, 0\}$, exhaustive permutation floors are coarse (min $p = 0.1667$ at K=3), and large fractions of the null mass tie at single values (75.4% at ρ=+0.5 for the Vṛtra–Theogony ladder). Only the K=5 pair is adequately powered at $\alpha = 0.05$, and even it has a p-floor of ~0.008. K=3 results in Table 3 — including the eye-catching sumerian–greek ρ=1.0 — are structurally unpowered and must not be read as evidence.

Construct and chronology caveats. The "magical" theme compares different text-type constructs across slots (Sumerian bilingual incantation tablets vs Hittite SISKUR ritual prescriptions), and the Sumerian magical documents are Late-Babylonian-period bilinguals, chronologically later than the mainly Old Babylonian ETCSL literary corpus. Both facts are recorded in the results JSON and bound the interpretation of any ladder that includes the magical theme.

Single-experiment scope. The retirement verdict binds the stronger-anchors lever on the semi-orthogonal Procrustes plane as operationalized here; it does not claim that no anchor improvement of any kind could ever matter elsewhere, and Hittite's low 0.0586/0.0666 cosine remains an open observation rather than an explained one.

5 Reproducibility

All computed artifacts — trained FastText models, fused vectors, Ridge/Procrustes weights, English embedding caches, processed corpora, anchors, eval results, and production exports (~45GB) — are mirrored in a companion Hugging Face dataset repo, **ebrinz/hyper-glyphy-artifacts**, whose layout mirrors this repository exactly. A fresh clone bolts on in one step:

```bash
[] bash shared/scripts/fetch_artifacts.sh # hf download + recreate local symlinks
```

This reproduces the published numbers exactly without retraining (FastText training is non-deterministic, so a retrain gives slightly different vectors). The pre-v2 artifact state is pinned as the `suite-v1` tag on the HF repo, from which the archived v1 table reproduces. The exact environment that produced the committed artifacts is pinned in `requirements.lock.txt` (Python 3.12.3; gensim 4.4.0, numpy 2.3.5, scikit-learn 1.7.0); the gensim FastText `.model` artifacts are version-sensitive pickles and must be loaded under the lockfile environment. Raw third-party corpora are not mirrored (licensing varies by source); each slot documents its fetch steps in `languages/<slot>/data/raw/README.md`. MW-derived Sanskrit files in the mirror carry CC BY-NC-SA 3.0 (Cologne CDSL); everything computed in the project is CC BY 4.0. Myth-study numbers in this paper are read directly from the git-tracked `shared/results/myth_study.json`; all verdict sentences are byte-copies from `docs/EXPERIMENT_JOURNAL.md` (entries 2026-07-16, 2026-07-19, 2026-07-24).

References

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- **ORACC** — The Open Richly Annotated Cuneiform Corpus (incl. ePSD2, DCCLT, blms). https://oracc.museum.upenn.edu/
- **TLA** — Thesaurus Linguae Aegyptiae (https://thesaurus-linguae-aegyptiae.de/); with Ramses (http://ramses.ulg.ac.be/) and BBAW (Berlin-Brandenburgische Akademie der Wissenschaften), via the predecessor heiroglyphy project.
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